

Experiment 'p_2_4_e_distance' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_e_distance

Natural language statement: null

Method signature: p_2_4_e_distance(x: int, y: int) returns (result: int)

Ensures

- `result >= 0`
- `if x >= y then result == x - y else result == y - x`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_e_distance(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result >= 0, ensures if x >= y then result == x - y else result == y - x`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2jo2lO9Em7iZD2hdE1P2vNKuDr7

Model: gpt-5.5-2026-04-23

Created at: 1780580968

Finish reason: stop

Usage: 286 tokens in, and 158 tokens out

```
{
  if x >= y {
    result := x - y;
  } else {
    result := y - x;
  }
}
```

Final Program

```
method p_2_4_e_distance(x: int, y: int) returns (result: int)
  ensures result >= 0
  ensures if x >= y then result == x - y else result == y - x
{
  if x >= y {
    result := x - y;
  } else {
    result := y - x;
  }
}
```

}

Total Token Usage

Input tokens: 286

Output tokens: 158

Reasoning tokens: 107

Sum of 'total tokens': 444

Experiment Timings

Overall Experiment started at 1780580966160, ended at 1780581002473, lasting 36313ms (36.31 seconds)

Iteration #1 started at 1780580966162, ended at 1780581002473, lasting 36311ms (36.31 seconds)